

OIS-011 — Conformance Verification

Status: Foundational

Authority: Implementation

Depends on:

- OIS-000 through OIS-010
 - OAS-000 through OAS-031
 - OCS-000 through OCS-032
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Purpose

This specification defines the implementation responsibilities for Constitutional Conformance Verification.

Verification determines whether an implementation faithfully realizes the Constitutional and Architectural Specifications.

Verification SHALL evaluate constitutional behavior.

Verification SHALL NOT evaluate implementation style.

Equivalent implementations MAY differ internally while remaining constitutionally conformant.

1. Implementation Objective

The objective of Conformance Verification is to establish observable, repeatable, implementation-independent methods for determining constitutional correctness.

Verification SHALL evaluate:

Constitutional preservation.

Architectural compliance.

Behavioral equivalence.

Historical continuity.

Operational reconstructability.

Verification SHALL remain technology-neutral.

2. Definition

Conformance Verification is the executable process through which an implementation demonstrates constitutional equivalence with the Organodynamics Specifications.

Verification SHALL evaluate behavior.

It SHALL not evaluate source code.

3. Verification Scope

Verification SHALL include:

Identity continuity.

Event preservation.

Schema compatibility.

Conversation continuity.

Knowledge integrity.

Decision integrity.

Federation compatibility.

Recovery correctness.

Architectural compliance.

Constitutional traceability.

Additional verification domains MAY be introduced without altering previous verification results.

4. Constitutional Traceability

Every verification result SHALL be traceable to one or more specifications.

Verification SHALL identify:

Applicable OCS specifications.

Applicable OAS specifications.

Applicable OIS specifications.

Verification SHALL remain independently reproducible.

5. Behavioral Equivalence

Implementations SHALL be considered equivalent when they preserve:

Constitutional meaning.

Architectural intent.

Observable behavior.

Historical reconstruction.

Behavioral equivalence SHALL take precedence over implementation similarity.

6. Verification Evidence

Verification SHALL produce explicit evidence.

Evidence MAY include:

Behavioral observations.

Event histories.

Identity mappings.

Recovery demonstrations.

Architectural analyses.

Operational traces.

Evidence SHALL remain historically attributable.

7. Recovery Verification

Verification SHALL demonstrate recovery after:

Implementation replacement.

Storage migration.

Network partition.

Partial failure.

Complete operational restart.

Recovery SHALL preserve constitutional continuity.

8. Federation Verification

Federated implementations SHALL demonstrate:

Independent operation.

Schema compatibility.

Identity compatibility.

Behavioral compatibility.

Recovery compatibility.

Federation SHALL preserve sovereignty.

9. Verification Levels

Verification MAY be performed at multiple levels.

Examples include:

Component.

Engine.

Place.

Federation.

Reference implementation.

Complete constitutional ecosystem.

Equivalent constitutional behavior SHALL remain the acceptance criterion at every level.

10. Verification Automation

Implementations SHOULD support automated verification.

Automation SHALL preserve:

Reproducibility.

Repeatability.

Traceability.

Historical comparison.

Automation SHALL never replace constitutional interpretation.

11. Certification

Communities MAY establish certification programs based on this specification.

Certification SHALL evaluate constitutional conformance.

Certification SHALL NOT prescribe implementation technologies.

Certification SHALL remain independently reproducible.

12. Conformance Verification Test

Conformance Verification satisfies this specification when independent verification teams, using independent verification implementations, consistently conclude that independently developed systems preserve equivalent constitutional behavior despite differing technologies, architectures, and engineering practices.

Conformance

An implementation conforms to this specification when constitutional meaning, architectural intent, historical continuity, operational behavior, and reconstructability are demonstrably preserved through independent, repeatable, technology-neutral verification.

Implementation Principle

Software can claim compliance.

Only verification can demonstrate it.

The purpose of Conformance Verification is therefore not to judge implementations,

but to protect the Constitution by ensuring that every implementation,

regardless of language,

vendor,

organization,

or technology,

can prove that it preserves the same constitutional understanding.

End of Specification OIS-011